Microsoft Elevate

CAPSTONE PROJECT

AI-Based Job Market Trend Analysis & Skill Recommendation System

PRESENTED BY

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OUTLINE:

- **Problem Statement** (Should not include solution)
- **Proposed System/Solution**
- **System Development Approach** (Technology Used)
- **Algorithm & Deployment**
- **Result (Output Image)**
- **Conclusion**
- **Future Scope**
- **References**

PROBLEM STATEMENT:

In today's rapidly evolving job market, students and job seekers face difficulty in identifying the most in-demand skills and trending job roles. Due to the vast availability of job postings across multiple platforms, manually analyzing job market trends is time-consuming and inefficient. There is no intelligent system that provides real-time insights into trending skills and personalized recommendations based on actual job market data. As a result, many students learn outdated or irrelevant skills, which affects their employability and career growth.

PROPOSED SOLUTION:

- The proposed system aims to develop an **AI-based Job Market Trend Analysis and Skill Recommendation System** that analyzes real-time job market data using Natural Language Processing (NLP) and Machine Learning (ML) techniques.

Key Features:

- Collect job postings dataset from online job platforms.
- Perform data preprocessing and NLP-based skill extraction.
- Analyze job descriptions to identify trending skills.
- Recommend personalized skills based on selected job roles.
- Provide real-time skill suggestions through a web-based interface.
- This system helps students, job seekers, and career counselors make data-driven career decisions.

SYSTEM APPROACH:

- **System Requirements:**
- Hardware:
  - Processor: Intel i3 or higher
  - RAM: 4GB minimum
  - Storage: 20GB free space
- Software:
  - Python
  - Flask Framework
  - VS Code
  - Google Colab
  - Web Browser
- **Libraries Used:**
- pandas
- scikit-learn
- nltk
- flask
- matplotlib

ALGORITHM & DEPLOYMENT:

Algorithm Selection:

- The system uses **Natural Language Processing (NLP)** techniques for text processing and **frequency-based learning models** for identifying trending skills. Machine Learning techniques are applied to recommend relevant skills for each job role.

Data Input:

- Job Title
- Job Description
- Industry
- Location

Training Process:

- Data cleaning and preprocessing
- Tokenization and text normalization
- Stopword removal
- Skill keyword extraction
- Frequency-based pattern learning

Prediction Process:

- User inputs job role
- System analyzes similar job descriptions
- Extracts frequent skills
- Displays top recommended skills

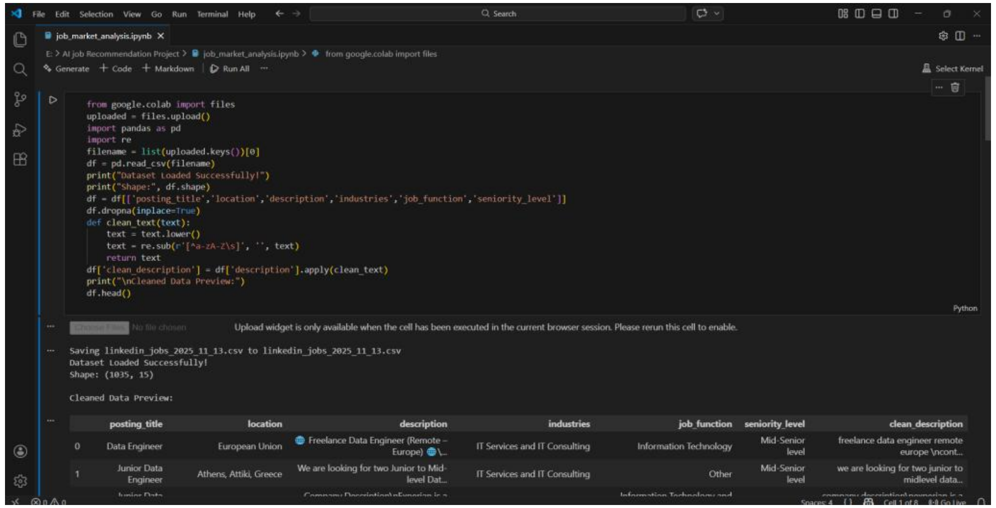
Deployment:

- The system is deployed using **Flask web framework** as a local web application, allowing users to interact via a browser interface.

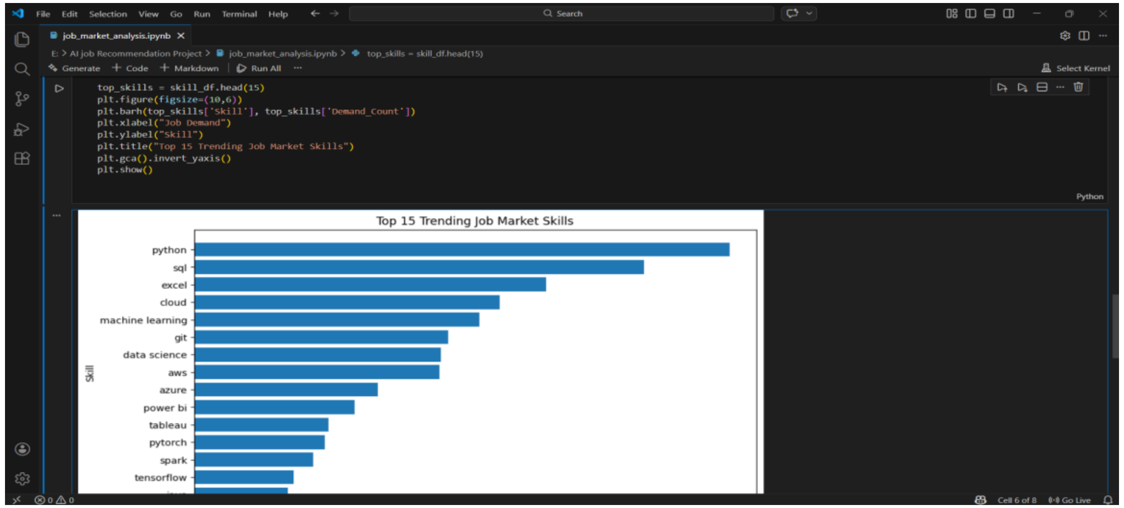
SYSTEM ARCHITECTURE

- User Input → Web Interface → Flask Server → NLP Processing → Skill Extraction → AI Recommendation Engine → Output Display

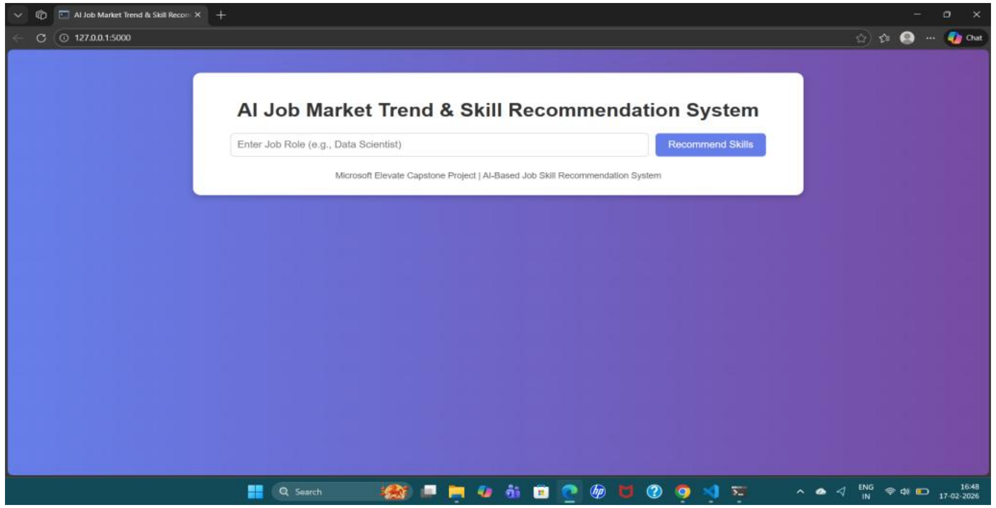
RESULT:



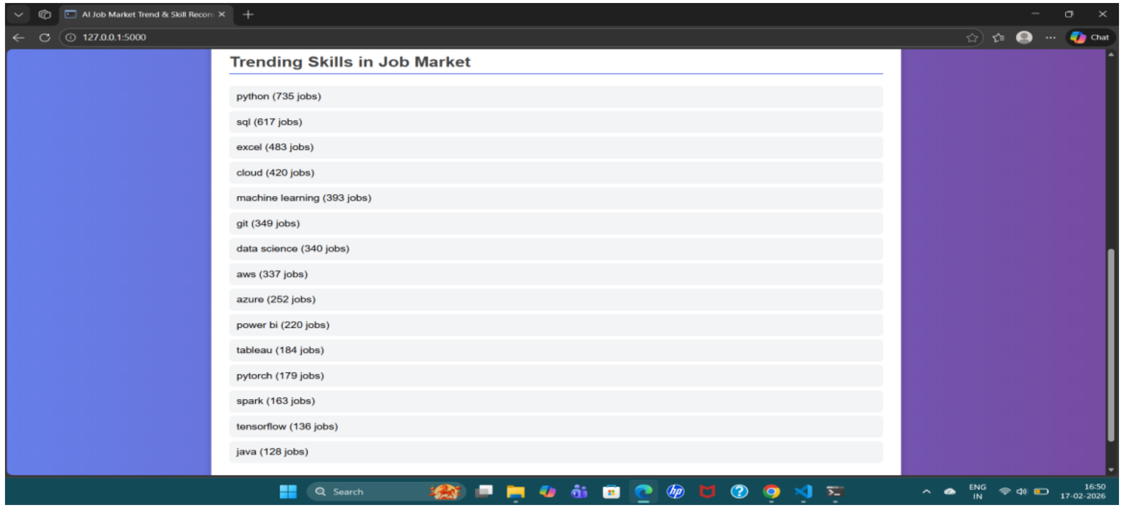
job_market_analysis.ipynb
from google.colab import files
uploaded = files.upload()
import pandas as pd
import re
filename = list(uploaded.keys())[0]
df = pd.read_csv(filename)
print("Dataset Loaded Successfully!")
print("Shape:", df.shape)
df = df[['posting_title','location','description','industries','job_function','seniority_level']]
df.dropna(inplace=True)
def clean_text(text):
text = text.lower()
text = re.sub(r'[^a-zA-Z\s]', '', text)
return text
df['clean_description'] = df['description'].apply(clean_text)
print("\nCleaned Data Preview:")
df.head()
Upload widget is only available when the cell has been executed in the current browser session. Please rerun this cell to enable.
Saving linkedin_jobs_2025_11_13.csv to linkedin_jobs_2025_11_13.csv
Dataset Loaded Successfully!
Shape: (1035, 15)
Cleaned Data Preview:



job_market_analysis.ipynb
top_skills = skill_df.head(15)
plt.figure(figsize=(10,6))
plt.barh(top_skills['Skill'], top_skills['Demand_Count'])
plt.xlabel("Job Demand")
plt.ylabel("Skill")
plt.title("Top 15 Trending Job Market Skills")
plt.gca().invert_yaxis()
plt.show()
Top 15 Trending Job Market Skills
python
sql
excel
cloud
machine learning
git
data science
aws
azure
power bi
tableau
pytorch
spark
tensorflow
Skill



127.0.0.1:5000
AI Job Market Trend & Skill Recommendation System
Enter Job Role (e.g., Data Scientist)
Recommend Skills
Microsoft Elevate Capstone Project | AI-Based Job Skill Recommendation System



127.0.0.1:5000
Trending Skills in Job Market
python (735 jobs)
sql (617 jobs)
excel (483 jobs)
cloud (420 jobs)
machine learning (393 jobs)
git (349 jobs)
data science (340 jobs)
aws (337 jobs)
azure (252 jobs)
power bi (220 jobs)
tableau (184 jobs)
pytorch (179 jobs)
spark (163 jobs)
tensorflow (136 jobs)
java (128 jobs)

CONCLUSION:

The AI-Based Job Market Trend Analysis & Skill Recommendation System effectively analyzes real job market data and provides intelligent career guidance. By leveraging NLP and ML techniques, the system delivers accurate insights into trending skills and helps students and job seekers make informed career decisions. This project demonstrates the practical application of Artificial Intelligence in solving real-world problems and contributes positively to career development.

FUTURE SCOPE:

- Integration with **live job APIs** (LinkedIn, Indeed, Naukri)
- Advanced AI models such as **Deep Learning (BERT, LSTM)**
- Career path prediction
- Resume analysis and job matching
- Cloud deployment with public access

REFERENCES:

List and cite relevant sources, research papers, and articles that were instrumental in developing the proposed solution. This could include academic papers on bike demand prediction, machine learning algorithms, and best practices in data preprocessing and model evaluation.

GitHub Link: Link

Thank You